

Experiment Playbook

This document provides an in-depth explanation of A/B testing, including its methodology, the creation of an Experiment Brief, and the use of various calculators to determine sample size, test duration, and significance.

- A/B testing involves comparing two hypotheses, using a p-value to determine if observed effects are statistically significant, and considering errors and effect size for reliability.
- An Experiment Brief is essential for aligning teams on the test's purpose, changes, audience, and success metrics, typically created by the Product Manager.
- The Product Analyst plays a crucial role in selecting metrics, calculating sample size and duration, validating tracking, and setting up result monitoring.
- Calculators provided include functions for determining sample size, test duration, and statistical significance, with practical significance assessed for long-running tests.
- Practical significance evaluates whether observed improvements are meaningful for business decisions, even if statistical significance is not achieved

What's A/B testing?

A/B testing or Hypothesis testing is a statistical method used to decide whether a change, difference, or effect observed in data is real or just due to random chance.

You start with a **Null Hypothesis (H_0)** that assumes “no change” or “no effect,” and an **Alternative Hypothesis (H_1)** that claims there *is* an effect.

You collect sample data and compute a **test statistic** and a **p-value**, which tells you how likely it is to see your result if the null hypothesis were true.

- If $p \leq \alpha$ (usually 0.05), you **reject H_0** → evidence supports the effect.
- If $p > \alpha$, you **fail to reject H_0** → not enough evidence.

The process includes concepts like **Type I error** (false positive), **Type II error** (false negative), **statistical power**, and **effect size**, which help determine reliability and practical meaning of the result.

In short:

Hypothesis testing checks if a result is meaningful or just random noise.

Sample Size and Test Duration Calculator

A. Sample Size Calculator

The function `sample_size_calculator_multigroup()` calculates the sample size required for an A/B/n test using a proportion test for CVRs.

Parameters passed through the function:

- Baseline: Current average CVR (as a proportion, e.g., 0.15 for 15% or as a continuous, e.g. 2.3).
- Desired lift percentage: Desired relative increase in CVR (percentage).
- Groups: Total number of groups in the test (e.g., 3 for A/B/C).
- Significance level: Significance level (alpha), default is 0.05.
- Power: Power (1 - beta), default is 0.8.
- Two-sided: Boolean indicating if the test is two-sided.
- Test type: Type of test to use, either 'z' or 't'.

Returns:

- The output **sample_size_per_group** gives the **minimum number of users needed per group** to reliably detect the desired lift with the chosen significance level and power.

For test planning, this value should be compared with the **average eligible traffic per day** to estimate the **test duration**.

How to decide the 'Test Type'?

- Use the **test_type = 'z' only when the primary metric is a proportional metric** (e.g., CVR, CTR, subscription rate, save rate, click-through rate).
- Use the **test_type = 't' only when the primary metric is continuous** (e.g., Time Spent per User, Images per User)

Explanation of sample size calculator:

1. New Conversion rate (p_2)

p_2 is the new conversion rate after applying the uplift to the base line

$$p_2 = \text{baseline} * (1 + \text{desired_lift_percentage} / 100)$$

2. Calculate Cohen's h (effect size for proportions)

Cohen's h is a standardized effect size used to measure the difference between two proportions (conversion rates in A/B testing).

It tells us how big the difference is between the control group (p_1) and the treatment group (p_2) in a standardized way.

3. Adjust alpha based on the n groups

Bonferroni correction is used when there are multiple experimental groups (A/B/n), regardless of whether the primary metric is proportional or continuous. It protects from inflated false-positive (Type I errors) rates caused by multiple comparisons and therefore increases the required sample size as the number of groups increases.

It prevents inflated false positives in A/B/n tests.

Since testing multiple groups increases the chance of a false positive, Bonferroni adjusts the significance level (α) by dividing it by the number of comparisons ($\alpha_1 = \alpha/n$).

The More number of groups $\rightarrow$ Stricter $\alpha \rightarrow$ Larger sample size needed.

if it's a test between two groups,

4. Compute Z-scores or T-scores

- When **proportional metric** (e.g., CVR, CTR, subscription rate, save rate, click-through rate) is passed then as per the test_type= z the Z-scores would be calculated.
- When **continuous metric** (e.g., Time Spent per User, Images per User) is passed then as per the test_type= t the T-scores would be calculated.

5. Calculate the sample size using power analysis

- Higher alpha/beta scores $\rightarrow$ more samples needed,
- Smaller effect size $\rightarrow$ more samples needed t

B. Test Duration Calculator

It estimate the duration (in days) required to run an A/B/n test based on available daily traffic.

Parameters:

- sample_size_per_group: Required number of users per variant (from sample size calculation)
- groups: Total number of groups/variants in the experiment (e.g., 3 for A/B/C)
- daily_traffic: Average number of eligible users per day landing on the experiment page

Returns:

- Estimated number of days required to complete the test

Significance Calculator

`perform_ab_test()` : Perform A/B tests across buckets for continuous or proportional metrics.

Parameters:

- `df` : DataFrame
Input dataframe containing the experiment data and metric values for each user/session.
- `experiment_id` : str
Unique identifier for the experiment (used for labeling and reporting results).
- `metric` : str
The name of the primary metric column used to evaluate experiment performance (e.g., Time Spent per User, CVR, CTR, Save Rate).
- `metricA` : str, optional
Name of the metric column for the Control group.
Only required when `df` stores metrics separately for A and B groups.
If None, the function assumes metric values are in a single combined column.
- `metricB` : str, optional
Name of the metric column for the Variant group.
Only required when metrics are stored separately across groups.
- `metric_type` : str, default "continuous"
Defines the type of statistical test:
 - "continuous" → t-test (metrics like Time Spent per User, Images per User)
 - "proportional" → z-test (conversion metrics like CVR, CTR, Subscription Rate)

- `alpha` : float, default 0.10

Significance level of the hypothesis test.

Lower alpha → more strict test → fewer false positives → more sample required.

- `lift_threshold` : float, default 5

Minimum % lift required to consider Variant a meaningful improvement (protects against calling small/noisy improvements as wins).

- `power_threshold` : float, default 0.80

Minimum statistical power required to trust the test sensitivity.

Higher power requirement → more sample needed, but fewer false negatives.

Returns:

dict —

A dictionary containing:

- `control_mean` / `variant_mean`
- `absolute_lift` and `percent_lift`
- `p_value`
- `power_score`
- `decision` (Win / Lose / Inconclusive)

Practical Calculator

The Statistical Significance calculator **`perform_ab_test()`** is used to evaluate whether an A/B test should **continue or stop** when it has already been running for a substantial time (typically **4+ weeks**) and a decision is required.

Even if statistical significance is not achieved yet, sample size isn't big enough to call a winner. Therefore, **practical significance** should also be assessed to decide whether the observed improvement is meaningful for the business.

Why practical significance matters?

So when a test runs long (e.g., 4+ weeks) without a clear directional result, checking for **practical significance** is essential for a confident decision.

Statistical significance tells whether the observed **difference is real**.

Practical significance tells whether the difference is **worth acting on**.

compute_practical_sig(): Computes practical significance for A/B tests by evaluating confidence intervals and comparing lift against a practical threshold.

Confidence Interval Visualization for Control vs Variant

If you wish to plot a graph to check the sample distribution, you can use the code below. It visualizes the control and variant curves with their confidence intervals, helping you clearly compare mean differences in any A/B test.

plot_distribution():

This function helps visualize how the control and variant groups differ by plotting their sampling distributions and confidence intervals, making it easier to understand mean separation, uncertainty, and the overall impact of an A/B test.

It plots the probability density curves for both groups and highlights their 90% confidence intervals, making it easier to visually compare the separation between means and uncertainty ranges.

Parameters:

- `control_mean` (float): Mean value of the metric for the control group.
- `control_se` (float): Standard error of the control group mean.
- `variant_mean` (float): Mean value of the metric for the variant group.
- `variant_se` (float): Standard error of the variant group mean.
- `metric_name` (str): Name of the metric being visualized (default = "Metric").
- `alpha` (float): Significance level for the confidence interval (default = 0.10 for 90% CI).

How to use:

1. Compute the mean and standard error for both control and variant.
2. Call the function with these values—for example:

```
plot_distribution(  
    control_mean=4.65,  
    control_se=0.98,  
    variant_mean=7.67,  
    variant_se=1.24,  
    metric_name="BPU (Images Saved per Saver)"  
)
```

The output graph helps you:

- See how far apart the means are.
- Check the overlap (or separation) of confidence intervals.

- Understand whether the variant lifts appear meaningful visually.

Here are some available online tools to check the significance of a 2-variant A/B test.

Note: These calculators support proportional (conversion-based) metrics and only two buckets.

- [A/B-Test Calculator - Power & Significance - ABTestGuide.com](#)
- [Statistical Significance Calculator for A/B Testing](#)